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Lesson Notes

MODULE 6 | LESSON 3

FACTOR MODELS WITH MACHINE LEARNING

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the last lesson, we discussed Smart Beta and the risks that come with factor investing. In this lesson, we review several machine learning techniques and compare their performance at predicting returns.

1. Why Machine Learning for Factor Models?

The first required reading for this lesson, a paper by Gu et al. that forms the backbone of this lesson, aims to provide an overview and comparative analysis of machine learning methods for estimating risk premiums to use in factor models. It assumes your familiarity with the standard machine learning model and techniques, which you should have acquired in the Machine Learning course. In case you do not remember all the details of each of those techniques from that course, the other required reading for this lesson and the next one will serve as a refresher. As a financial engineer in the twenty-first century, it can only help you to know very well the main machine learning techniques and the theory that supports them.

Pay special attention to the three-part definition of machine learning that Gu et al. propose. Do you agree with this definition? Why or why not? What would you add, subtract, or change about this definition so that it is consistent with your understanding of machine learning?

Gu also discusses some reasons that machine learning (ML) is especially well suited to the challenge of factor model (FM) estimation (2225–6). For example, one reason is that FMs are really models for predicting asset prices: If asset prices are expected to increase, one should buy; if they are expected to decrease, one should sell. It just so happens that one of ML's specialties is prediction. As a second reason, we have already discussed the ever-growing multitude of factors and many of the recent additions have dubious value; luckily, ML entails several techniques for thinning this "zoo" of factors with various variable selection and dimension reduction techniques. A third reason that ML appears to be a smart choice for the FM challenge is that ML hardly

constitutes a single approach, but rather, it represents a very large family of approaches and techniques. So whatever the functional form of the FM under consideration, there is most likely an ML model that can help approximate it.

2. Which ML Techniques and Why?

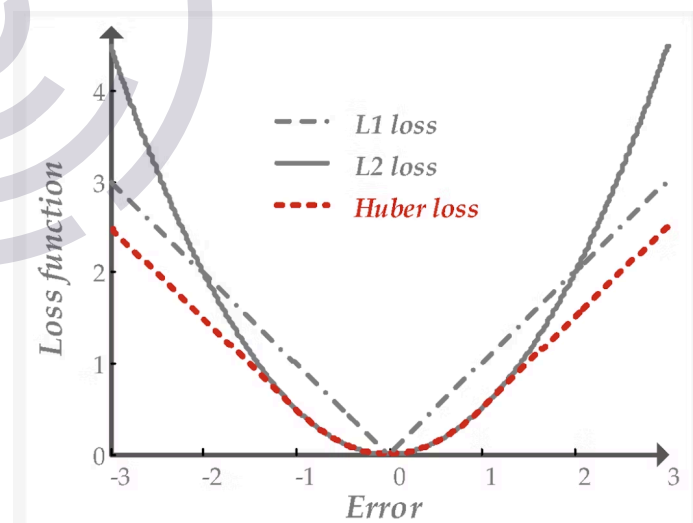
Given that there are so many models and techniques in play in the paper, make sure you understand the purpose of each one as it is mentioned. You should know, for example, the difference between

- Selection and shrinkage
- Training and validation and testing
- Lasso and ridge regression and elastic net
- Linear regression and generalized linear regression models
- Principal components regression and partial least squares regression
- Basic regression trees and boosted trees and random forests
- Layers, neurons and activation functions in a neural network

If you do not recall these basic differences, Gu et al.'s paper reviews them at a high level; in addition, the other required readings (chapters 5–10 of [this ebook](#) and the code for the relevant chapters, where available, from [this repo](#)) will support you. *Don't forget that you can always go back to the lesson notes from the Machine Learning course as well.*

Pay close attention to the ways that parsimony is enforced in the model construction process, e.g., the Huber robust objective function, the elastic net, etc. For example, Gu et al.'s study starts with simple linear models and reviews ordinary least squares (OLS), and then extends OLS. First, we see a generic weighted version of OLS, and then we are introduced to the Huber robust objective function: "a hybrid of squared loss for relatively small errors and absolute loss for relatively large errors" (Fang 2234).

Figure 1: Huber Loss Function in Relation to L1 and L2 Loss Function



Source: Wu, Weijiang, et al. "Benchmarking Daily Line Loss Rates of Low Voltage Transformer Regions in Power Grid Based on Robust Neural Network." *Applied Sciences*. vol. 9, no. 24, 2019, p. 5565. <https://www.mdpi.com/2076-3417/9/24/5565>.

You can see from this plot that the Huber loss function indeed shares characteristics of its L1 and L2 relatives. Remember that as important as knowing the specifics of this function is,

understanding why and how such modifications are made is also critical so that you are prepared to innovate when your research circumstances require it.

Make sure you understand the important advantages and disadvantages of all of the models. For example, know their differences regarding how they handle interactions between predictors, if at all. A more complex model may have certain advantages, such as incorporating such interactions, but at what cost? To what extent can you explain the parameter values in a neural net? Know the hyperparameters that the various models require and, of course, what these hyperparameters do and how they are chosen.

3. Performance

Gu et al.'s paper provides an insightful discussion of ML applications for FMs, but the comparative analysis is also quite valuable. In this section, understand how the authors decided to evaluate model performance in their comparison—and why. Is their reason for modifying the well-known R^2 calculation surprising to you?

How do the authors assess the importance of variables as predictors? If they use more than one method to make these assessments, do the different methods provide distinct insights? Do the different methods even agree?

These methodological questions aside, what are the results? What stock characteristics and macroeconomic variables have the most predictive power? If different models or approaches disagree about the importance of the variables, do the authors' explanations for these disagreements make sense? For example, were you surprised by the relative importance of overall market volatility and/or aggregate book-to-market ratio as predictors, and if you were surprised, why?

4. Conclusion

In this lesson, we reviewed the advantages and disadvantages of several machine learning techniques for the task of risk premium estimation. In the next lesson, we go further by building on the neural network suite of techniques to actually construct—not just find—the optimal factors for inclusion in our factor model.

References

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- Esposito, Vinicius. "Machine Learning for Factor Investing." *GitHub*. https://github.com/viniesposito/py-mlfactor/blob/main/py_mlfactor.ipynb.
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